

Visual Data Exploration- Due Diligence Project

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Variables, description and Type: Original dataset

1. CustomerID: Unique identifier for each customer. Type: Character.
2. Region: Region where the customer is located. Type: Character.
3. TownSize: Size of the town where the customer resides. Type: Character.
4. Gender: Gender of the customer. Type: Character.
5. Age: Age of the customer. Type: Character.
6. EducationYears: Number of years of education completed by the customer. Type: Character.
7. JobCategory: Category of the customer's job. Type: Character.
8. UnionMember: Indicates whether the customer is a member of a union. Type: Character.
9. EmploymentLength: Number of years the customer has been employed. Type: Character.
10. Retired: Indicates whether the customer is retired. Type: Character.
11. HHIncome: Household income. Type: Character.
12. DebtToIncomeRatio: Debt-to-income ratio of the customer. Type: Character.
13. CreditDebt: Amount of credit debt the customer has. Type: Character.
14. OtherDebt: Other types of debt the customer has. Type: Character.
15. X: Unknown variable. Type: Character.
16. LoanDefault: Indicates whether the customer defaulted on a loan. Type: Character.
17. MaritalStatus: Marital status of the customer. Type: Character.
18. HouseholdSize: Size of the customer's household. Type: Character.
19. NumberPets: Number of pets owned by the customer. Type: Character.
20. NumberCats: Number of cats owned by the customer. Type: Character.
21. NumberDogs: Number of dogs owned by the customer. Type: Character.
22. NumberBirds: Number of birds owned by the customer. Type: Character.
23. HomeOwner: Indicates whether the customer owns a home. Type: Character.
24. CarsOwned: Number of cars owned by the customer. Type: Character.
25. CarOwnership: Ownership status of the customer's car. Type: Character.
26. CarBrand: Brand of the customer's car. Type: Character.
27. CarValue: Value of the customer's car. Type: Character.
28. CommuteTime: Commute time for the customer. Type: Character.
29. PoliticalPartyMem: Indicates whether the customer is a member of a political party. Type: Character.
30. Votes: Indicates whether the customer votes. Type: Character.
31. CreditCard: Type of credit card used by the customer. Type: Character.

32. CardTenure: Tenure of the customer's credit card. Type: Character.
33. CardItemsMonthly: Number of items purchased with the customer's credit card monthly. Type: Character.
34. CardSpendMonth: Amount spent with the customer's credit card monthly. Type: Character.
35. ActiveLifestyle: Indicates whether the customer has an active lifestyle. Type: Character.
36. PhoneCoTenure: Tenure with the customer's phone company. Type: Character.
37. VoiceLastMonth: Amount spent on voice services last month. Type: Character.
38. VoiceOverTenure: Tenure with the customer's voice service provider. Type: Character.
39. EquipmentRental: Indicates whether the customer rents equipment. Type: Character.
40. EquipmentLastMonth: Amount spent on equipment rental last month. Type: Character.
41. EquipmentOverTenure: Tenure with the customer's equipment rental provider. Type: Character.
42. CallingCard: Indicates whether the customer uses a calling card. Type: Character.
43. WirelessData: Indicates whether the customer uses wireless data services. Type: Character.
44. DataLastMonth: Amount spent on data services last month. Type: Character.
45. DataOverTenure: Tenure with the customer's data service provider. Type: Character.
46. Multiline: Indicates whether the customer has multiline service. Type: Character.
47. VM: Indicates whether the customer uses voicemail. Type: Character.
48. Pager: Indicates whether the customer uses a pager. Type: Character.
49. Internet: Type or level of internet service subscribed to by the customer. Type: Character.
50. CallerID: Indicates whether the customer has caller ID service. Type: Character.
51. CallWait: Indicates whether the customer uses call waiting. Type: Character.
52. CallForward: Indicates whether the customer uses call forwarding. Type: Character.
53. ThreeWayCalling: Indicates whether the customer uses three-way calling. Type: Character.
54. EBilling: Indicates whether the customer uses electronic billing. Type: Character.
55. TVWatchingHours: Number of hours spent watching TV by the customer. Type: Character.
56. OwnsPC: Indicates whether the customer owns a personal computer. Type: Character.
57. OwnsMobileDevice: Indicates whether the customer owns a mobile device. Type: Character.
58. OwnsGameSystem: Indicates whether the customer owns a game system. Type: Character.
59. OwnsFax: Indicates whether the customer owns a fax machine. Type: Character.
60. NewsSubscriber: Indicates whether the customer subscribes to a news service. Type: Character.

Feature engineering performed for the dataset:

1. Removed the **customerID** column from the dataset. These data entries serve no purpose at this point. These cannot be used to find any significant correlation with another variable and is an identifier.

2. The **HHIncome** variable has entries with "\$" and "," and is a character variable so I removed the "\$" and "," and turned the character variable into numerical so it can be graphed into a histogram to show the frequency and possibly put into a range. These values were later logged and compared with the log of the **TotalDebt** variable (CreditDebt + OtherDebt variable). The extreme outliers are also removed which have skewed the distribution quite a bit.
3. **CreditDebt** and **OtherDebt** variables are expressed in \$100,000 so the values are in decimal points. We can multiply these values by 100,000 and then use that for analysis to give a true output that is easier to understand. This allows easier graphing and makes interpretation easier.
4. Creation of a new variable when CreditDebt and OtherDebt are added together (**TotalDebt**). This is expressed as a double dbl variable or num character to allow easier analysis with another numerical variable.
5. **X variable** is removed. It is a blank variable and is most likely added due to the merging of two datasets.
6. **CarOwnership** and **CarBrand** variables have -1 entries if it is not Own/Lease or Domestic/Foreign so if entries are anything but those are now changed to other. The -1 prohibits graphing of those variables in R and the reason for -1 entry is not stated thus can be changed to other.
7. **CarValue** and **CarSpendMonth** variables have entries with "\$" and even "(" which in R creates errors that are removed. Then the values are turned into numerical variables from character variables. Again this makes graphing easier on R and regression plots or scatter plots can be created.
8. **VoiceLastMonth** and **VoiceOverTenure** also have entries that are "\$", and "," EquipmentLastMonth and EquipmentOverTenure have "\$" and "-" entries which are all removed and if NaN values are created, those are omitted when doing analysis. When NaN values are present, for these kinds of entries it is hard to impute values, and adding mean values would change the entire distribution, and getting true meaning would be difficult. So I left them to others and then plotted a histogram so the count could be used for analysis.
9. **DataOverTenure** and **DataLastMonth** also have similar entries.
10. Internet variables have a yes, no, 2,3, or 4 so if the entry is anything but Yes or No is changed to Other. This will allow graphing to see the frequency of Yes and No and others.
11. The **Gender Variable** has missing entries and those were changed to others. Similar reasoning to number 10.
12. **Log transformation** of the HHincome variable, along with the addition of CreditDebt and OtherDebt variables to create a TotalDebt variable which was later logged. At first, the

graph had a cluster near the 0s of x and y, and doing log transformation spread out the points showing a proper correlation between the two variables.

Provide several examples of qualitative or graphical analysis of features

Single variable plots (chosen by the correct type of data - categorical, ordinal, real-values)

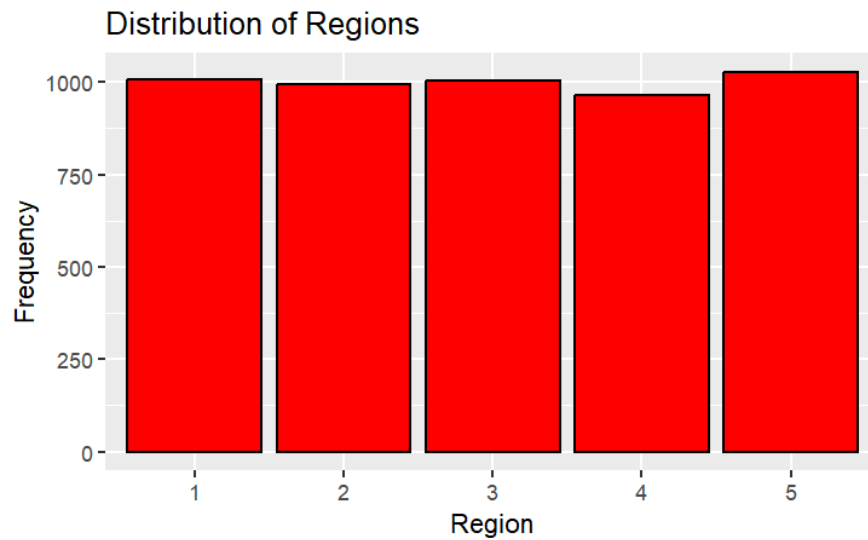


Figure 1: Distribution of customer regions represented by numerical codes, ranging from 1 for NorthEast to 5 for Southeast.

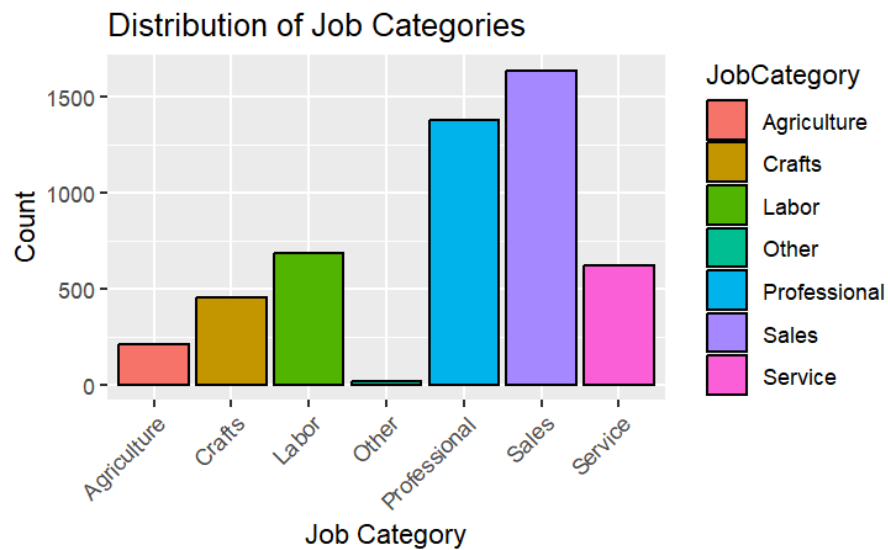


Figure 2: Distribution of Job Categories

The figure displays the distribution of job categories among customers, categorized into various sectors or natures of employment. It shows that sales had the highest occurrence and Agriculture and Others were the lowest.

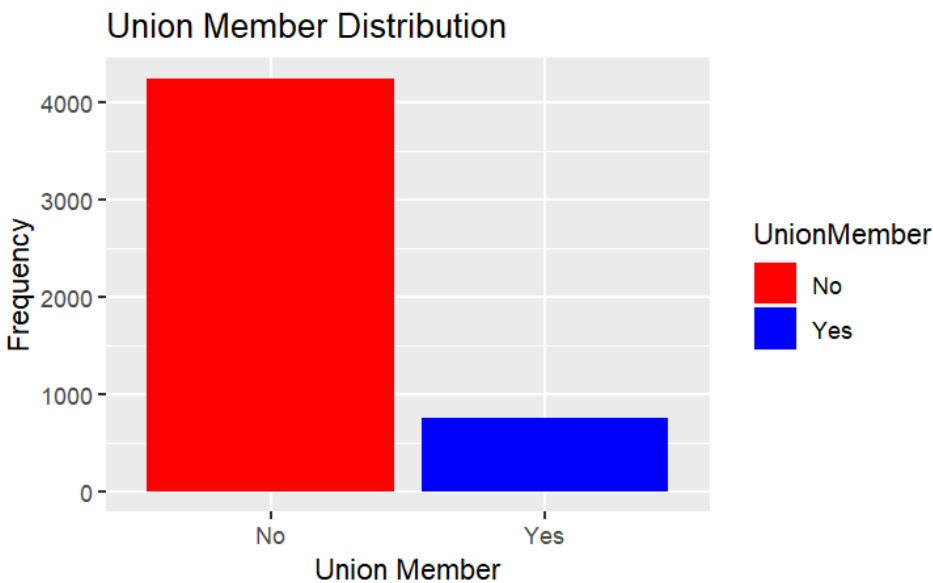


Figure 3: Distribution of Union Members

The figure displays the distribution of union member among the customers, separated into yes and no.

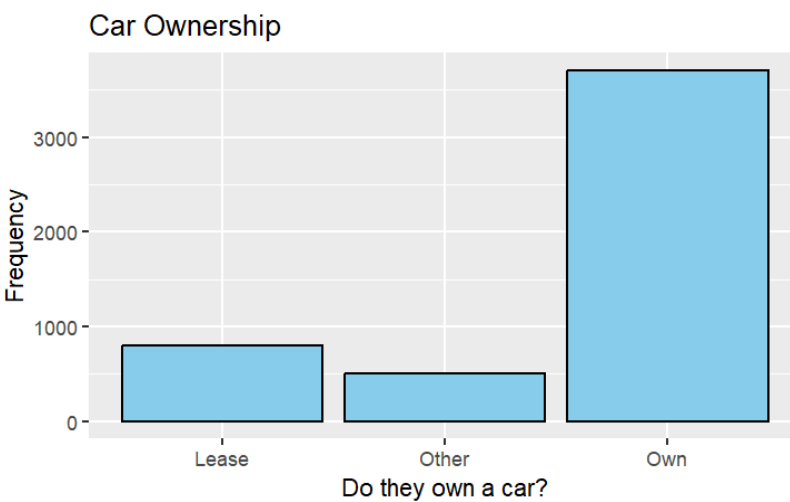
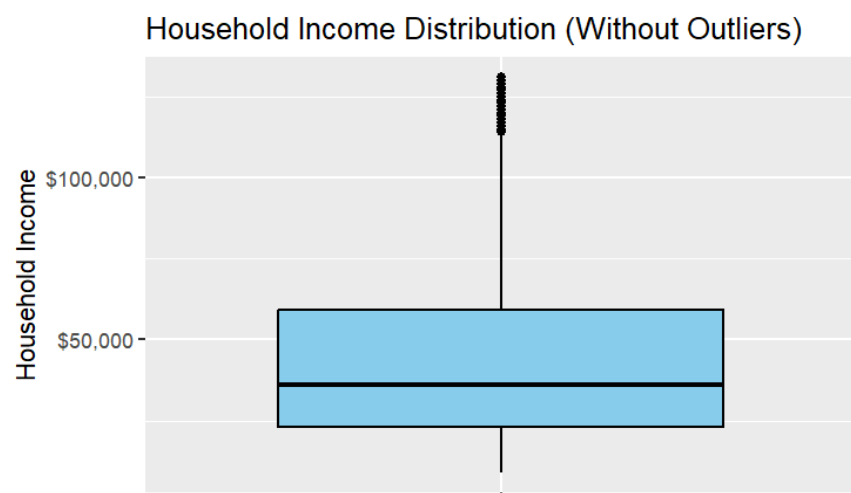
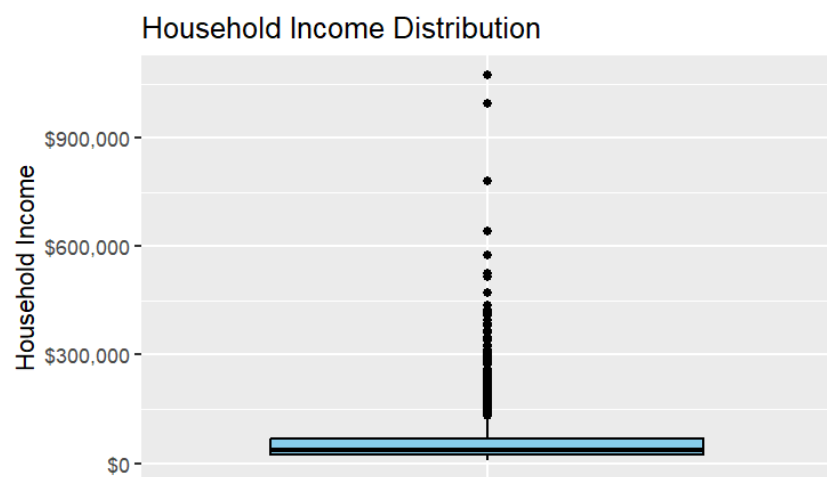
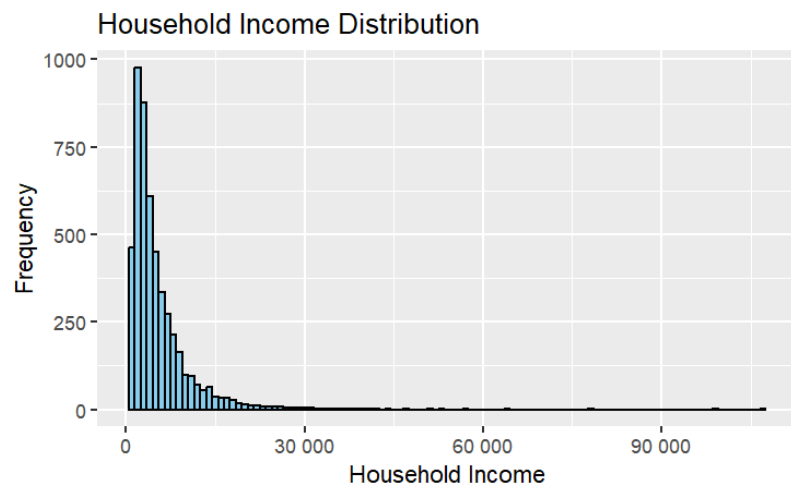


Figure 4: Distribution of Car Ownership

The figure displays the distribution of Car Ownership among the customers, separated into Lease, Own, and others. Others were missing entries or nonapplicable. Most customers own their vehicle.



Figures 5, 6, and 7.

The graphs above show the distribution of household income of customers, then a box plot and another box plot with extreme outliers removed.

Perform at least 5 two-column analyses (scatterplots, regression fits)



Figure 8 Scatterplot of Age Vs. Employment Length.



Figure 9. Box plot for Household income depending on gender.

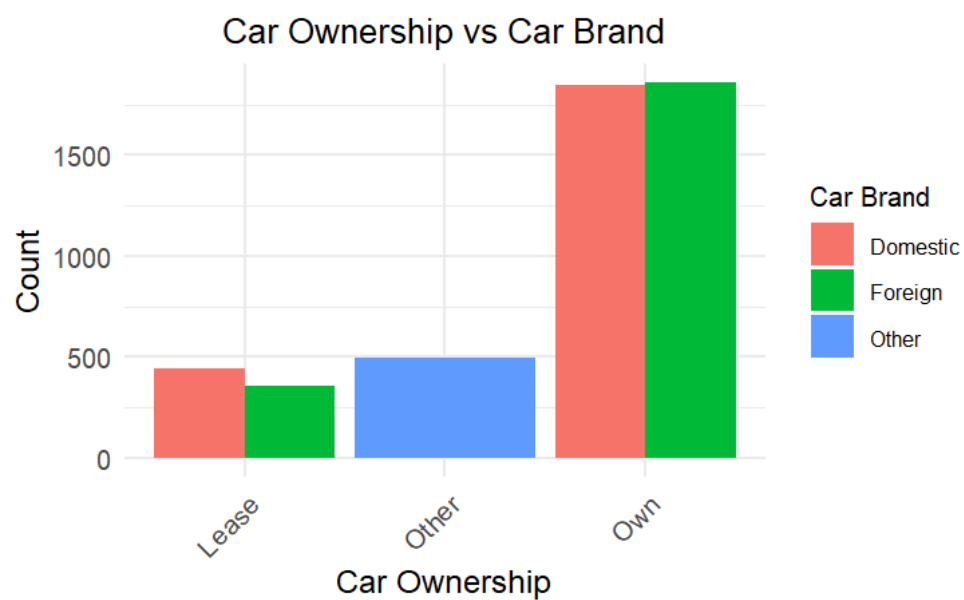


Figure 10. Barplot for Car ownership against Car Brand.

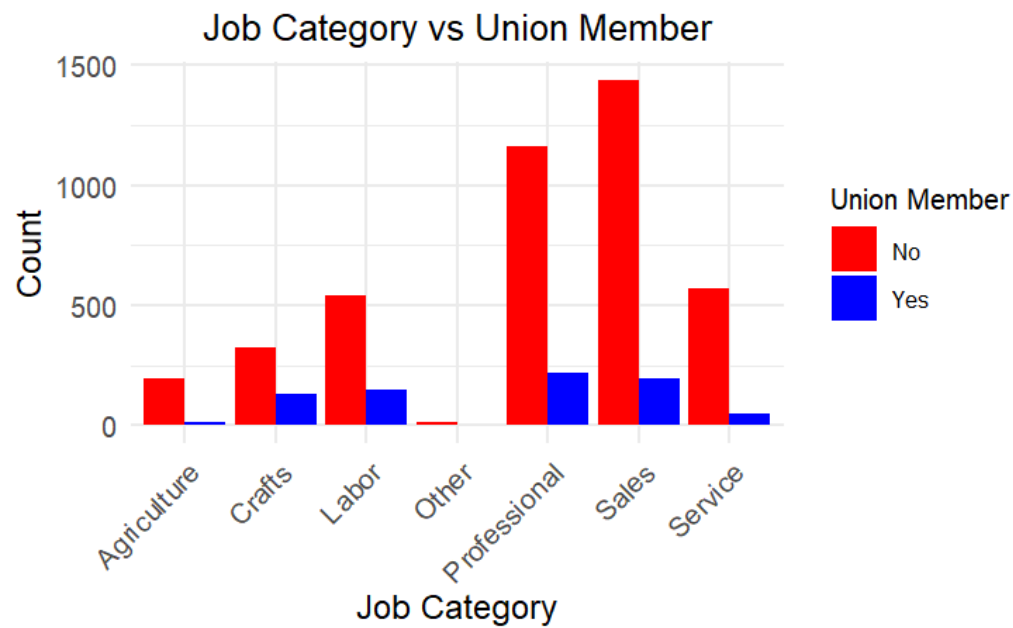


Figure 11. Barplot of Job category vs union member.

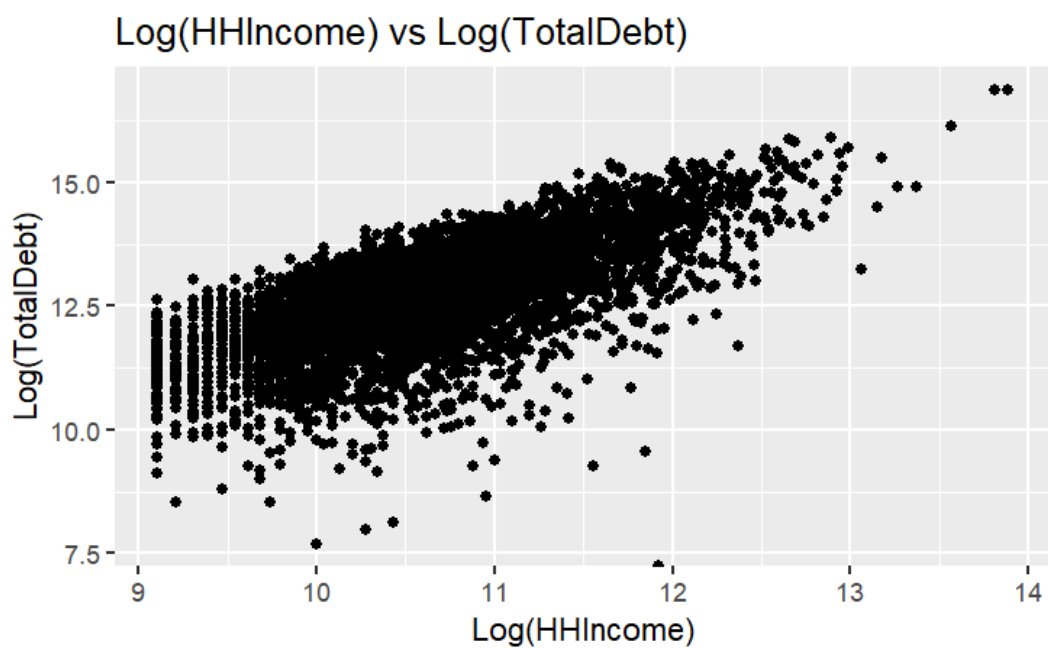


Figure 12. Scatter plot of logged values of Total Debt and Household income.